Über die Daubechies Wavelets

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The Daubechies $2N$ wavelet ($Daub(2N)$) is an orthonormal wavelet that is smooth and has compact support of width $2N$.

Let $\phi \in L^2(\mathbb{R})$ be a scaling function of $Daub(2N)$, that is,

$$\widehat{\phi}(2\xi) = m_0(\xi)\widehat{\phi}(\xi)$$

where m_0 is a lowpass mask which is a trigonometric polynomial.

To construct the lowpass mask (denoted by τ or m_0) of $Daub(2N)$ satisfying the so-called "Cohen condition", namely

$$|m_0(\xi)|^2 + |m_0(\xi+\pi)|^2 = 1, \quad \xi \in \mathbb{R},$$

we start from the basic identity:

$$\left(\cos^2\left(\frac{\xi}{2}\right) + \sin^2\left(\frac{\xi}{2}\right)^2\right)^{2N-1} = 1.$$

We group the half terms with higher powers of $\cos^2(\xi/2)$ to express $|m_0|^2$ as follows:

$$\begin{aligned}
|m_0(\xi)|^2 &= \sum_{n=0}^{N-1} \binom{2N-1}{n} \sin^{2n}\left(\frac{\xi}{2}\right)\cos^{2(2N-1-n)}\left(\frac{\xi}{2}\right) \\
&= \cos^{2N}\left(\frac{\xi}{2}\right)\left[\sum_{n=0}^{N-1} \binom{2N-1}{n} \sin^{2n}\left(\frac{\xi}{2}\right)\cos^{2(N-1-n)}\left(\frac{\xi}{2}\right)\right] \\
&=: \cos^{2N}\left(\frac{\xi}{2}\right) r(\xi).
\end{aligned}$$

1 Regularity

Theorem 1.1. Let $N \geq 1$ and ϕ be a scaling function of $Daub(2N)$ with a lowpass mask m_0 satisfying

$$|m_0(\xi)|^2 = \cos^{2N}\left(\frac{\xi}{2}\right) r(\xi)$$

where $r(\xi) = \sum_{n=0}^{N-1} \binom{2N-1}{n} \sin^{2n}(\xi/2)\cos^{2(N-1-n)}(\xi/2)$. Define the operator T_r on $C[0,\pi]$ by

$$T_r u(\xi) = r(\xi/2)u(\xi/2) + r(\pi - \xi/2)u(\pi - \xi/2).$$

Then $\phi \in W^{s,2}$ if and only if $s < s^* = N - \log_4 \rho(T_r)$, where $\rho(T_r)$ is the spectral radius of T_r.

To obtain the optimal Sobolev degree, we need to compute the **spectral radius** $\rho(T_r)$ of T_r.

If the polynomial r admits a cosine expansion, i.e., $r(\xi) = \sum_{k=0}^{d-1} r_k \cos(k\xi)$, then T_r maps the space spanned by $\{\cos(k\xi)\}_{k=0}^{d-1}$ into itself. Hence, the spectral radius of T_r can be obtained from the eigenvalues of the resulting $d \times d$ matrix representation of T_r.

Remark. For the Daubechies wavelets $Daub(2N)$, the associated trigonometric polynomial r indeed has a cosine expansion.

Since $\sin^2(\xi/2) = (1-\cos\xi)/2$ and $\cos^2(\xi/2) = (1+\cos\xi)/2$, we obtain

$$r(\xi) = \sum_{n=0}^{N-1} \binom{2N-1}{n} \left(\frac{1-\cos\xi}{2}\right)^n \left(\frac{1+\cos\xi}{2}\right)^{N-1-n} = P(\cos\xi)$$

for some polynomial P of degree $\leq N-1$.

Since each power $\cos^m \xi$ can be written in a linear combination of $\cos(k\xi)$, $0 \leq k \leq m$, the trigonometric polynomial r can be expressed as

$$r(\xi) = \sum_{k=0}^{N-1} r_k \cos(k\xi)$$

for suitable coefficients r_k. Thus, the spectral radius of T_r can be computed from the eigenvalues of an $N \times N$ matrix.

Theorem 1.2. Let $N \geq 1$ and ϕ be a scaling function of $Daub(2N)$ with a lowpass mask m_0 satisfying

$$|m_0(\xi)|^2 = \cos^{2N}\left(\frac{\xi}{2}\right) r(\xi)$$

where $r(\xi) = \sum_{n=0}^{N-1} r_k \cos(k\xi)$ for some coefficients r_k. Define the operator T_r on the space spanned by $\{\cos(n\xi)\}_{k=0}^{N-1}$ by

$$T_r u(\xi) = r(\xi/2)u(\xi/2) + r(\pi - \xi/2)u(\pi - \xi/2).$$

Let A_r be the matrix representation of T_r with respect to the basis $\{\cos(n\xi)\}_{n=0}^{N-1}$; that is,

$$A_r := \left[\left[T_r(1)\right]_{\{\cos(n\xi)\}} \quad \cdots \quad \left[T_r(\cos(N-1)\xi)\right]_{\{\cos(n\xi)\}} \right].$$

Then the scaling function satisfies $\phi \in W^{s,2}$ if and only if $s < s^* = N - \log_4 \rho(A_r)$ where $\rho(A_r)$ denotes the spectral radius of A_r.

1.1 Example

For $N = 2$ (in case of $Daub4$), we have

$$\begin{aligned} |m_0(\xi)|^2 &= \cos^4(\xi/2)\left(\cos^2(\xi/2) + 3\sin^2(\xi/2)\right) \\ &= \cos^4(\xi/2)(2 - \cos\xi). \end{aligned}$$

The operator T_r is defined by

$$(T_r u)(\xi) = r(\xi/2)u(\xi/2) + r(\pi - \xi/2)u(\pi - \xi/2).$$

Here, with our r, we have

$$r(\xi/2) = 2 - \cos(\xi/2), \quad r(\pi - \xi/2) = 2 + \cos(\xi/2).$$

Using the basis $\{1, \cos\xi\}$, we compute the matrix representation A of T_r as follows:

(i) For $u(\xi) = 1$,

$$\begin{aligned} (T_r 1)(\xi) &= r(\xi/2) \cdot 1 + r(\pi - \xi/2) \cdot 1 \\ &= (2 - \cos(\xi/2)) + (2 + \cos(\xi/2)) \\ &= 4 \\ &= 4 \cdot 1 + 0 \cdot \cos\xi. \end{aligned}$$

(ii) For $u(\xi) = \cos\xi$,

$$\begin{aligned}(T_r \cos\xi)(\xi) &= r(\xi/2)\cos(\xi/2) + r(\pi - \xi/2)\cos(\pi - \xi/2) \\ &= (2 - \cos(\xi/2))\cos(\xi/2) + (2 + \cos(\xi/2))(-\cos(\xi/2)) \\ &= -2\cos^2(\xi/2) \\ &= -1 - \cos\xi \\ &= (-1)\cdot 1 + (-1)\cdot\cos\xi.\end{aligned}$$

This implies

$$A = \begin{bmatrix} T_r(1) & T_r(\cos\xi) \end{bmatrix} = \begin{bmatrix} 4 & -1 \\ 0 & -1 \end{bmatrix}.$$

Therefore, the eigenvalues of A are 4 and -1, which implies the spectral radius $\rho(T_r) = \max(|4|, |-1|) = 4$. Finally, we have

$$s^* = N - \log_4 \rho(T_r) = 2 - \log_4 4 = 1.$$

TABLE I
The Best Sobolev Regularity Exponents Found

	Sobolev regularity exponent s_0				
$2N$	$n_z = 0$	$n_z = 1$	$n_z = 2$	$n_z = 3$	$n_z = 4$
2	0.50				
4	1.00				
6	1.42	1.00			
8	1.78	1.82			
10	2.10	2.26	1.00		
12	2.39	2.66	2.00		
14	2.66	3.02	3.00	1.00	
16	2.91	3.37	3.48	2.00	
18	3.16	3.72	3.92	3.00	1.00
20	3.40	4.07	4.32	4.00	2.00
22	3.64	4.42	4.73	4.76	3.00
24	3.87	4.78	5.14	5.22	4.00
26	4.11	5.14	5.55	5.67	5.00
28	4.34	5.50	5.95	6.10	5.84
30	4.57	5.85	6.34	6.51	6.38
32	4.79	6.19	6.71	6.89	6.88
34	5.02	6.52	7.09	7.25	7.30
36	5.24	6.83	7.45	7.62	7.69
38	5.47	7.15	7.81	7.99	8.08
40	5.69	7.46	8.17	8.37	8.51